

RENSJON, Sweden

WMO: 020310

Lat: 68.073N

Lon: 19.835E

Elev: 1617

StdP: 13.86

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.5	-19.6	-30.5	1.0	-24.6	-25.1	1.4	-19.4	31.1	23.5	27.6	22.7	0.7	190	0.790

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	72.8	58.5	68.7	57.1	64.9	55.2	60.9	69.4	58.8	66.0	56.8	62.6	8.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.6	75.1	63.0	55.6	70.0	61.6	53.7	65.1	59.8	27.8	69.7	26.3	66.1	25.0	62.5	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	25.0	21.0	17.9	DB	-32.6	76.9	3.8	3.6	-35.3	79.5	-37.5	81.6	-39.7	83.6	-42.4	86.2
(5)				WB	-31.9	63.5	4.4	2.1	-35.1	65.0	-37.7	66.2	-40.2	67.4	-43.4	68.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.6	8.0	8.4	15.9	26.3	38.5	48.0	54.6	50.9	42.1	28.9	17.5	13.8
	DBStd	18.87	13.83	13.93	10.73	8.06	7.21	6.56	5.82	5.60	5.95	8.56	10.83	13.05
	HDD50	7784	1301	1166	1057	710	364	116	21	57	242	653	975	1122
	HDD65	12945	1766	1586	1522	1160	820	511	326	438	686	1118	1425	1587
	CDD50	318	0	0	0	0	9	55	163	84	7	0	0	0
	CDD65	4	0	0	0	0	0	0	3	1	0	0	0	0
	CDH74	72	0	0	0	0	3	12	50	7	0	0	0	0
	CDH80	6	0	0	0	0	0	1	5	0	0	0	0	0
Wind	WSAvg	6.9	6.3	6.5	7.9	6.9	7.5	7.9	7.0	6.3	7.1	6.5	5.8	6.6
Precipitation	PrecAvg	18.0	0.9	0.9	0.7	0.9	1.4	1.9	3.1	2.5	1.7	1.7	1.2	1.0
	PrecMax	25.2	2.0	1.5	1.7	1.9	3.7	3.9	8.9	5.3	4.1	4.5	2.3	2.5
	PrecMin	11.3	0.1	0.1	0.3	0.2	0.0	0.0	1.3	0.6	0.5	0.7	0.3	0.4
	PrecStd	4.1	0.5	0.5	0.4	0.5	1.0	1.2	1.9	1.2	1.0	0.9	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.8	41.6	40.6	50.8	71.5	74.9	79.4	74.3	62.8	51.0	44.1	42.0
		MCWB	34.7	36.4	35.1	42.0	53.1	57.1	62.2	60.6	53.9	45.5	39.2	37.3
	2%	DB	35.0	36.4	37.1	44.3	62.8	68.1	74.6	69.7	57.7	47.1	38.3	37.3
		MCWB	32.1	32.5	33.0	38.3	49.4	53.7	60.1	58.8	50.9	43.2	35.6	33.5
	5%	DB	30.7	32.4	34.4	41.2	56.2	63.8	70.3	65.2	54.5	43.7	34.6	33.4
		MCWB	28.3	29.6	31.0	36.1	46.7	52.6	58.7	56.9	49.2	40.8	32.6	30.7
	10%	DB	26.9	28.5	31.4	38.5	50.5	60.1	66.3	61.5	52.1	40.3	31.6	30.5
		MCWB	25.1	26.3	28.4	34.3	43.1	50.8	56.9	54.9	48.1	38.1	30.1	28.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.1	36.8	35.7	42.3	54.0	59.9	64.4	63.8	54.8	46.3	39.6	37.6
		MCDB	37.5	41.2	39.9	49.9	68.0	69.8	73.5	72.0	59.6	49.9	43.3	41.3
	2%	WB	32.0	32.6	33.4	38.8	50.5	56.6	61.9	60.0	52.2	43.4	35.9	34.0
		MCDB	34.8	36.2	36.5	43.6	62.7	63.4	71.7	66.8	55.9	46.7	37.9	36.8
	5%	WB	28.4	29.6	31.1	36.6	46.9	54.3	60.0	57.8	50.4	41.0	32.7	30.7
		MCDB	30.4	32.4	34.1	40.5	55.4	62.4	68.2	64.2	53.3	43.8	34.4	33.1
	10%	WB	25.1	26.3	28.6	34.6	43.8	51.7	58.1	55.7	48.6	38.3	30.3	28.3
		MCDB	26.9	28.2	31.3	38.1	50.2	59.0	64.5	60.6	51.4	40.1	31.5	30.0
Mean Daily Temperature Range	5% DB	MDBR	16.6	17.4	18.1	16.4	14.0	14.6	15.8	15.6	13.3	12.0	14.4	15.6
		MCDBR	16.2	16.3	16.7	16.9	25.0	21.9	23.2	22.9	18.3	14.6	14.5	14.9
		MCWBR	14.2	13.7	14.0	11.7	13.5	10.4	10.8	13.0	12.0	10.9	12.5	12.6
	5% WB	MCDBR	16.0	16.1	16.3	15.5	24.0	19.7	20.0	20.7	14.8	13.9	13.7	14.9
		MCWBR	14.4	13.7	13.9	10.7	13.4	10.5	10.3	12.4	10.4	10.5	12.0	12.9
Clear-Sky Solar Irradiance	taub	0.165	0.225	0.265	0.292	0.311	0.316	0.323	0.310	0.280	0.243	0.170	N/A	
	taud	1.932	2.144	2.235	2.276	2.435	2.519	2.526	2.539	2.584	2.323	1.977	N/A	
	Ebn at Noon	43	191	248	273	281	283	276	268	251	184	37	0	
	Edh at Noon	3	14	23	30	29	28	27	24	17	11	2	0	
All-Sky Solar Radiation	RadAvg	11	145	598	1218	1577	1572	1454	1053	563	186	30	0	
	RadStd	1	15	35	57	123	128	141	92	58	26	3	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page